



Review Test Submission: Assignment Week 6

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Course EBN 122 S2 2017

Test Assignment Week 6

Started 9/1/17 8:50 AM

Submitted 9/4/17 4:15 PM

Due Date 9/4/17 11:59 PM

Status Completed

Attempt Score 12.3 out of 28 points

Time Elapsed 79 hours, 24 minutes

Instructions

Read over the document *ClickUP Assignments Instructions*.

Before sending an enquiring regarding ClickUP question to any-one please verify that your problem is valid ... i.e. if you suspect incorrect marking first check with a fellow student or two.

Please direct any enquiries regarding ClickUP assignments to EBN2017.ClickUptest@gmail.com

Save the answer to each question as you complete it.

At the end of the assignment click Save and Submit.

Give your final correct answer 3 significant digits.

Make sure you use the FULL STOP for your decimal place marker.

You must indicate the units, unless otherwise specified:

- There must be NO space between the answer and the units
- There must be between 1 and 3 digits to the left of the decimal point. *This means that you need enter 1kV rather than 1000V and 100mV rather than 0.1V*
- Write out 'ohm' to indicate Ω (use singular)
- Use 'u' to indicate micro.
- Only use pico(p), nano(n), micro(u), milli(m), kilo(k), mega(M), giga(G) and tera(T); do not use centi, deci and deca

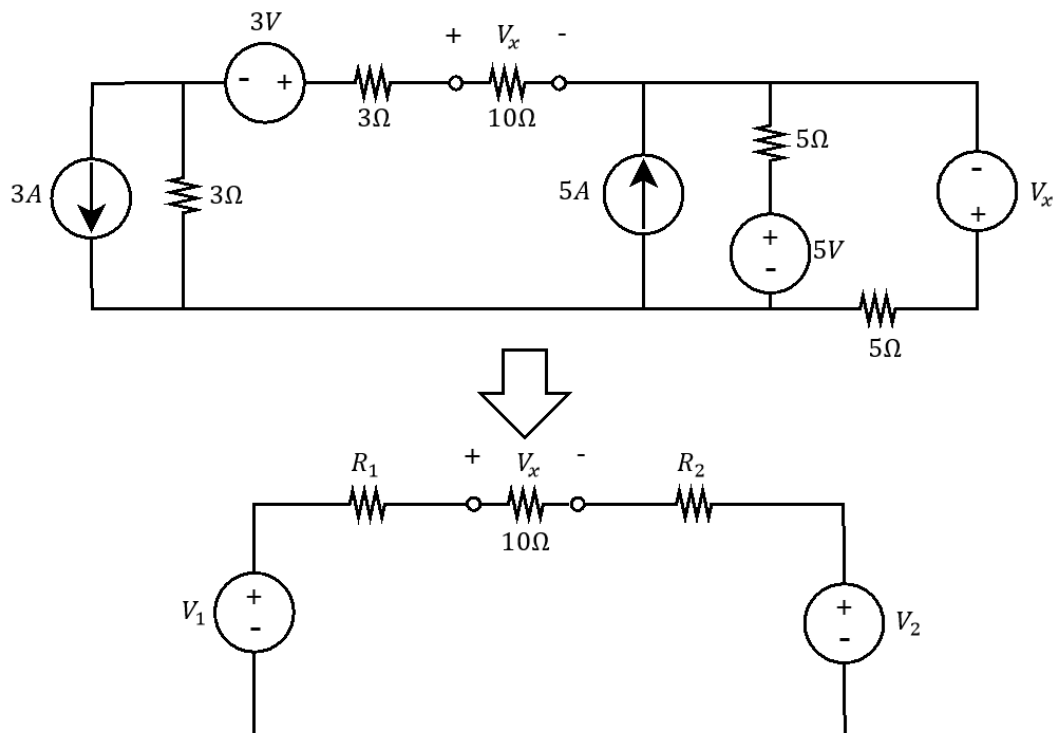
Results All Answers, Submitted Answers, Correct Answers, Feedback, Incorrectly Answered Questions

Question 1

2 out of 2 points



Use source transformation to simplify the circuit below for $V_x = 2$ V. Given: $V_1 = -6$ V, $R_1 = 6$ Ω , $R_2 = 2.5$ Ω , Find V_2 .



Selected Answer: ☒ 14V

Correct Answer: ☒ 14.00 ± 5% (V)

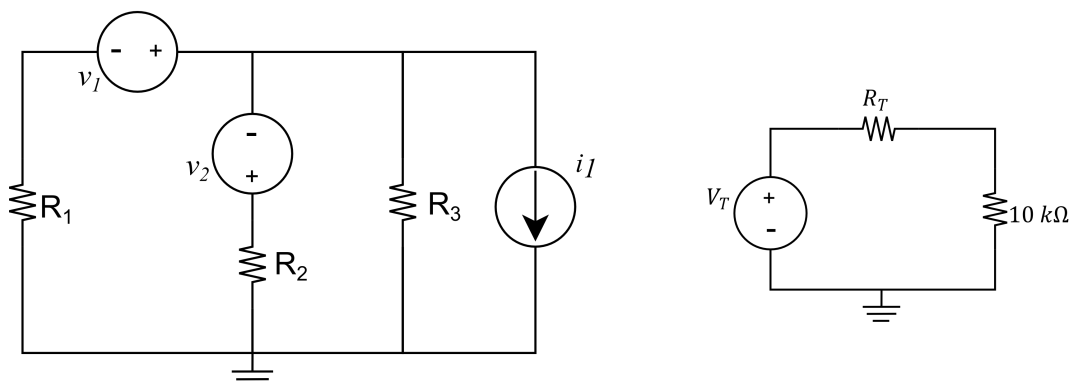
Question 2

0.5 out of 2 points



The aim is to simplify the circuit on the left to obtain the circuit on the right with respect to R_3 (which is $10\text{k}\Omega$). Make use of source transformation and determine R_T .

$R_1=8\text{k}\Omega$, $R_2=14\text{k}\Omega$, $V_1=14\text{V}$, $V_2=13\text{V}$ and $I_1=31\text{mA}$



Selected Answer: ☒ 22kohm

Correct Answer: ☒ 5.091 ± 5% (kohm)

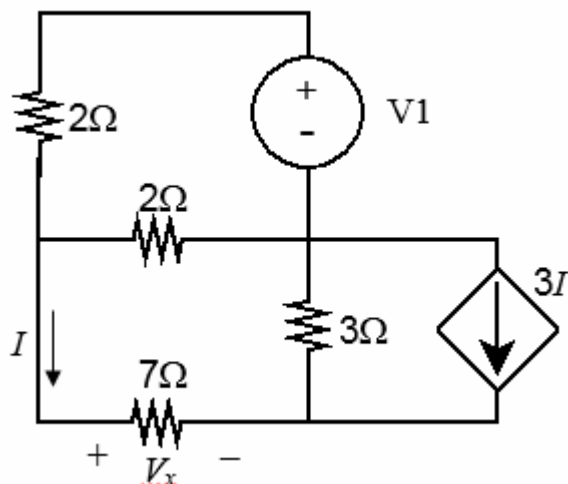
Question 3

2 out of 2 points



Use source transformation to determine V_x .

$V_1 = 11\text{ V}$.



Selected Answer: ☒ 1.93V

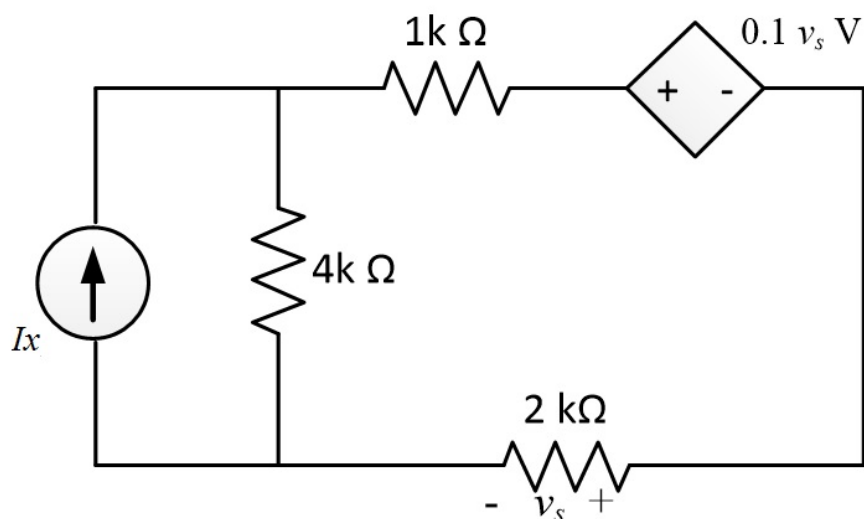
Correct Answer: ☒ $1.93 \pm 5\%$ (V)

Question 4

2 out of 2 points



If $I_x = 2$ mA, use source transformation to determine v_s



Selected Answer: ☒ 2.22V

Correct Answer: ☒ $2.22 \pm 5\%$ (V)

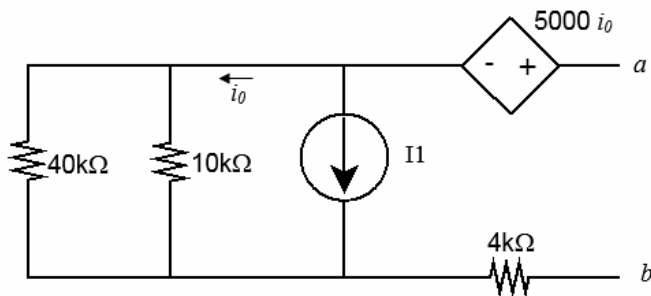
Question 5

0.2 out of 2 points



Determine the Thevenin voltage with regards to the terminals a and b.

$I_1 = 9$ mA



Selected Answer: ✗ -27V

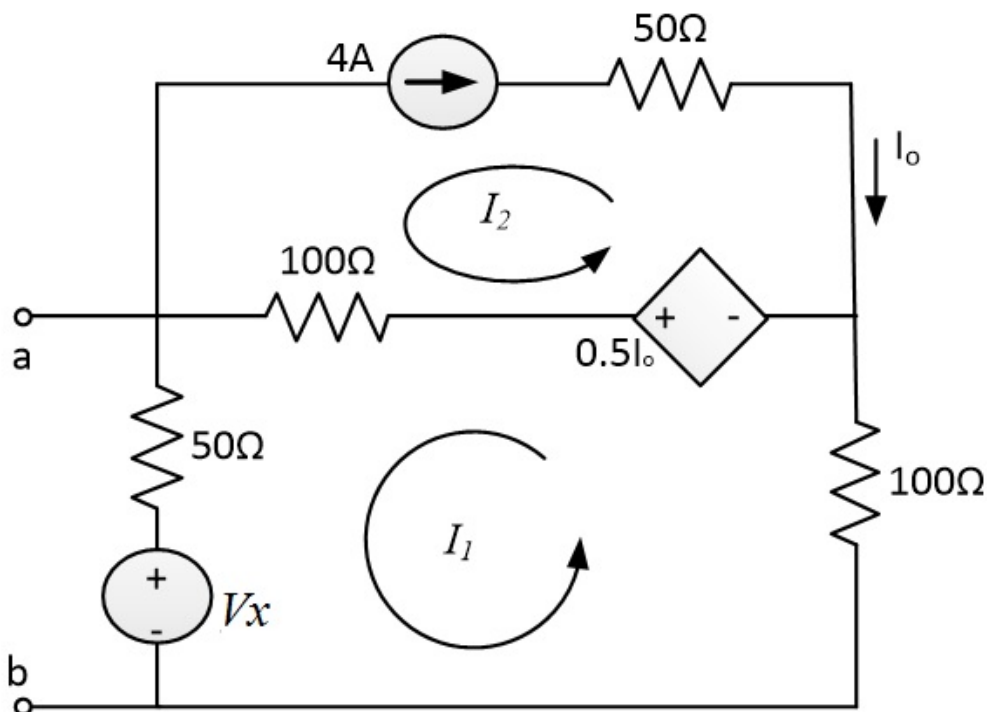
Correct Answer: ✓ -117.00 ± 2% (V)

Question 6

0.2 out of 2 points



Find the Thevenin equivalent voltage between terminals a-b if $V_x = 336$.



Selected Answer: ✗ 798V

Correct Answer: ✓ 189.20 ± 5% (V)

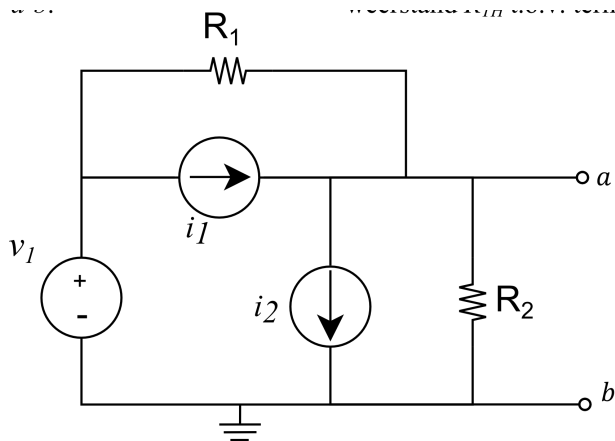
Question 7

2 out of 2 points



Determine the Thevenin equivalent voltage V_{TH} w.r.t. terminals a-b.

Let $R_1 = 5 \text{ k}\Omega$, $R_2 = 6 \text{ k}\Omega$, $v_1 = 12 \text{ V}$, $i_1 = 4 \text{ mA}$ and $i_2 = 2 \text{ mA}$.



Selected Answer: ☒ 12V

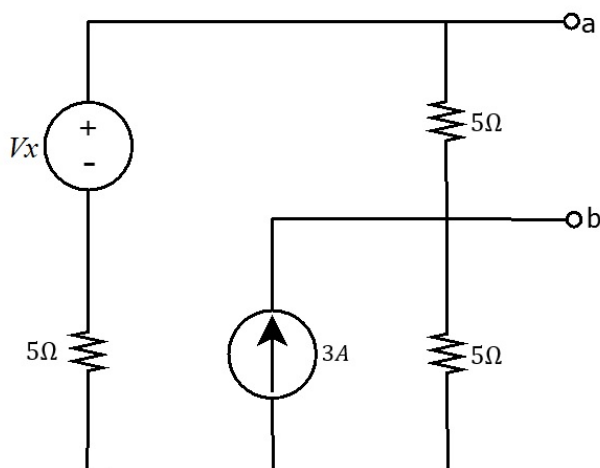
Correct Answer: ☒ 12.00 ± 5% (V)

Question 8

1.5 out of 2 points



Determine the equivalent Thevenin voltage at terminals a-b if $V_x = 22.76$



Selected Answer: ☒ 2.59

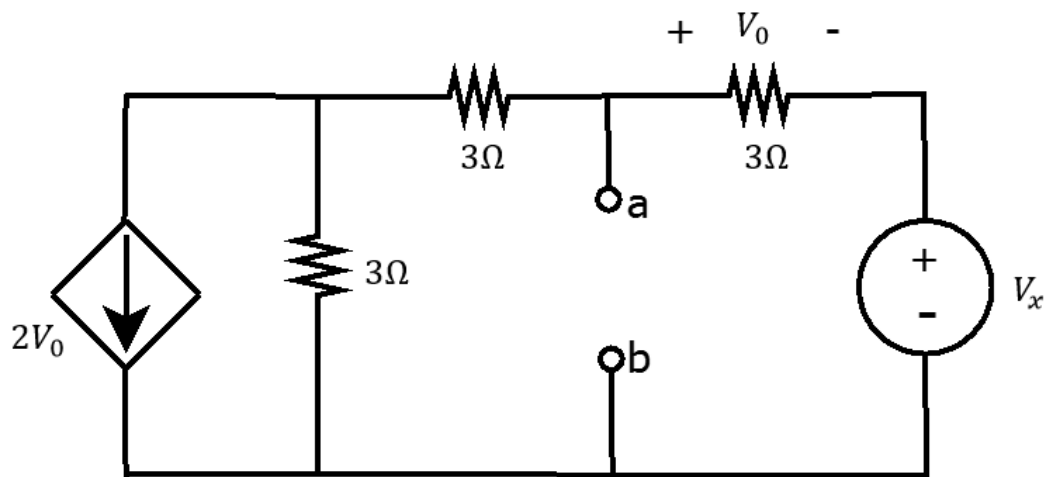
Correct Answer: ☒ 2.59 ± 5% (V)

Question 9

0 out of 2 points



Determine the Thevenin resistance wrt terminals a-b, given $V_x = 9.5$.



Selected Answer: ✗ 9ohm

Correct Answer: ✓ 666.67 ± 5% (mohm)

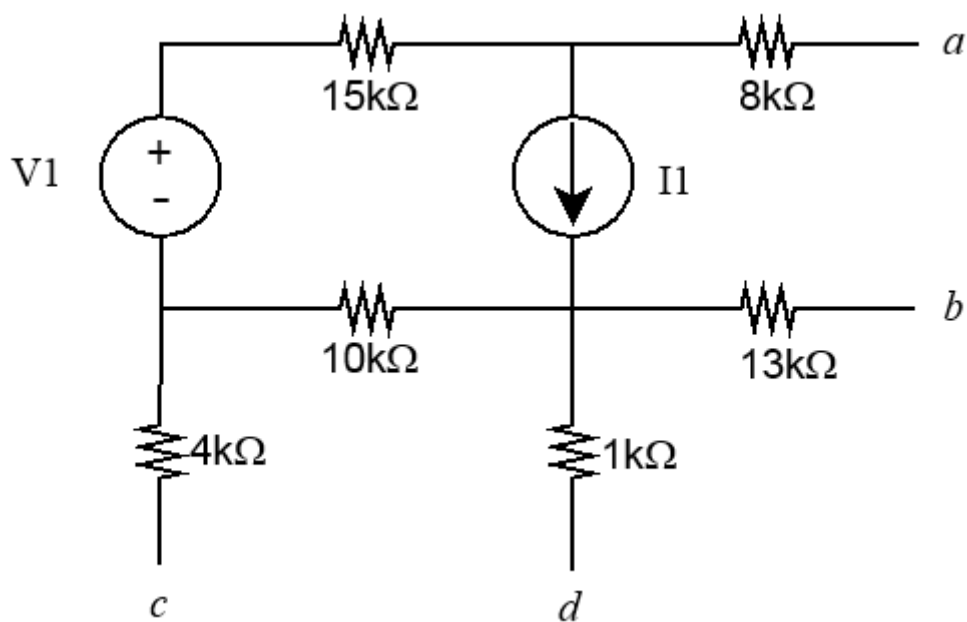
Question 10

0.2 out of 2 points



Determine the Thevenin equivalent resistance ($k\Omega$) in terms of terminals c and d.

$V_1 = 20\text{ V}$, $I_1 = 6\text{ mA}$.



Selected Answer: ✗ 12.8kohm

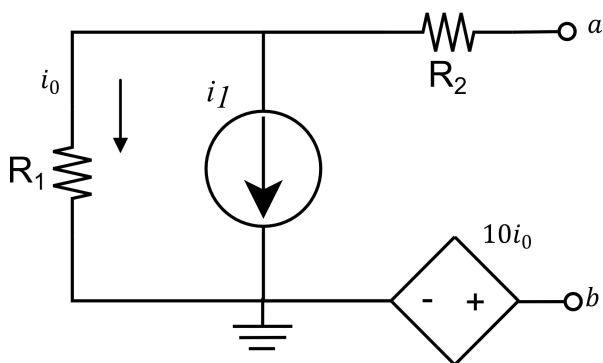
Correct Answer: ✓ 15.00 ± 5% (kohm)

Question 11

0.5 out of 2 points



Let $R_1 = 9$, $R_2 = 6$ and $i_1 = 19$. Determine the Norton equivalent current I_N



Selected Answer: ✗ 0.95A

Correct Answer: ✓ $3.80 \pm 5\%$ (A)

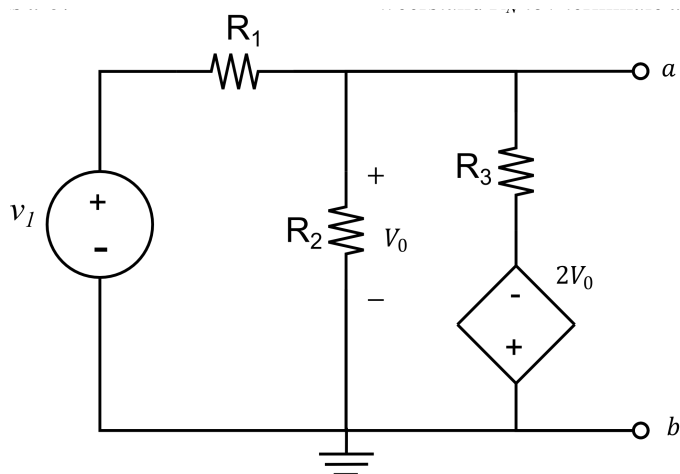
Question 12

0.5 out of 2 points



Determine the Norton equivalent current I_N w.r.t. terminals $a-b$.

$R_1=8k\Omega$, $R_2=6k\Omega$, $R_3=69k\Omega$ and $V_1=61V$



Selected Answer: ✗ 3.64mA

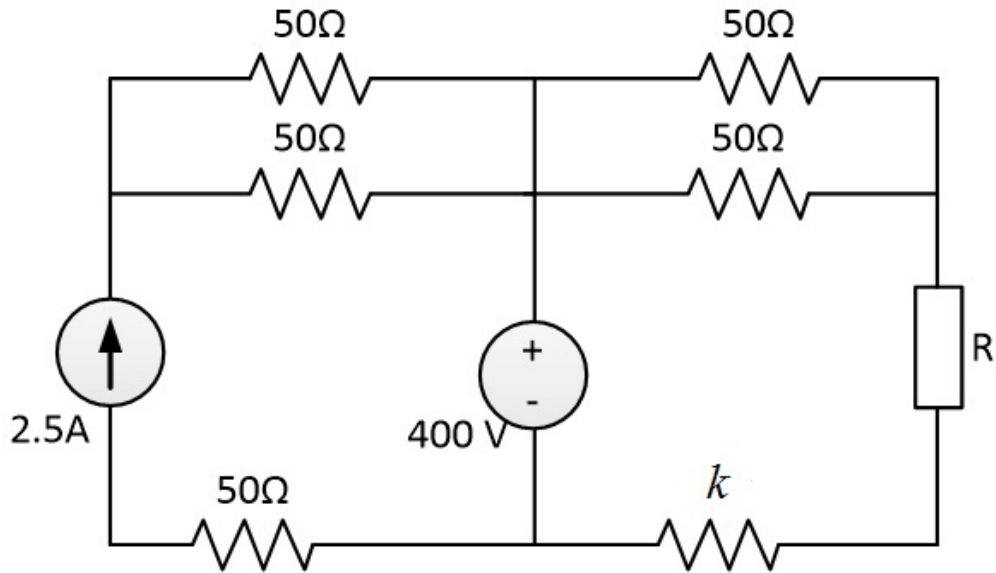
Correct Answer: ✓ $7.63 \pm 5\%$ (mA)

Question 13

0.2 out of 2 points



Find the equivalent Norton resistance with respect to the load resistor R if $k = 38$ ohms.



Selected Answer: ✗ 163ohm

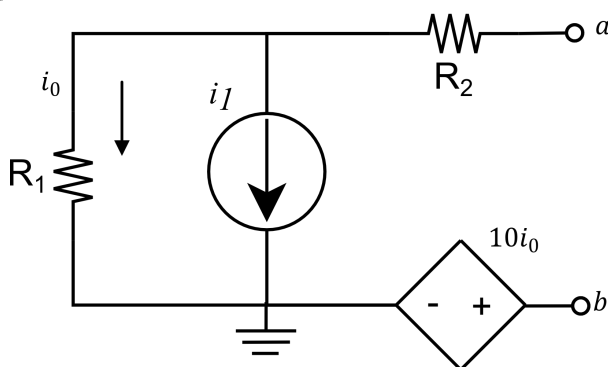
Correct Answer: ✓ $63.00 \pm 5\%$ (ohm)

Question 14

0.5 out of 2 points



Let $R_1 = 13$, $R_2 = 4$ and $i_I = 7$. Determine the resistance R_N w.r.t. terminals a-b. .



Selected Answer: ✗ 17ohm

Correct Answer: ✓ $7.000 \pm 5\%$ (ohm)

Monday, September 4, 2017 4:15:30 PM SAST

← OK